



# Low Reynolds Number Experimental Aerodynamic Verification of Scaled and LEWICE Simulated Ice Accretions in SLD Conditions

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An aerodynamic verification of the recommended scaling method by Anderson and Tsao and the LEWICE software was performed for ice formations generated underneath Duper-cooled Large Drop (SLD) cloud conditions specified under Title 14 of the Code of Federal Regulations (CFR), Part 25, Appendix O envelope. All experiments were performed at the University of Dayton Low Speed Wind Tunnel (UD-LSWT). Due to the large influence of the stagnation freezing fraction on ice shapes geometry, two different stagnation freezing fractions ( $n_o = 0.3$  and  $n_o = 0.5$ ) were investigated. Particle Image Velocimetry (PIV) was performed on the leading edge and near wake regions of the iced airfoils to evaluate the flow dynamics of the ice accretions. Modal analysis of the flow fields was performed using Proper Orthogonal Decomposition (POD) to evaluate the presence of coherent structures in the flow and their corresponding length scales. The results of the low Reynolds number aerodynamic investigation showed that ice accretions gathered on the scaled model inconsistently replicate the aerodynamics created by the ice accretions formed on the full-scale model for the given conditions. The stagnation freezing fraction was the primary difference between the scaled ice accretions, therefore, suggesting a possible stagnation freezing fraction constraint of the scaling method's use for SLD conditions. The LEWICE software consistently predicted an ice shape that was aerodynamically uncharacteristic of the full-scale ice accretion, therefore failing aerodynamic verification for the SLD conditions studied.

## I. Nomenclature

### Abbreviations:

AOA – Angle of Attack  
CFR – Code of Federal Regulations  
DPIV – Digital Particle Image Velocimetry  
FAA – Federal Aviation Administration  
FOV – Field of view  
GRC – Glenn Research Center  
IRT – Icing Research Tunnel  
NACA – National Advisory Committee for Aeronautics  
NASA – National Aeronautics and Space Administration  
PIV – Particle Image Velocimetry  
POD – Proper Orthogonal Decomposition  
SLD – Super-cooled Large Drop  
UD-LSWT – University of Dayton Low Speed Wind Tunnel

### Common Variables:

$A_C$  – Accumulation parameter, dimensionless  
 $AR$  – Aspect ratio, dimensionless  
 $b$  – Relative heat factor, dimensionless  
 $c$  – Airfoil chord, m

$C_d$  – Two-dimensional coefficient of drag, dimensionless  
 $C_l$  – Two-dimensional coefficient of lift, dimensionless  
 $C_{l,\alpha}$  – Two-dimensional theoretical coefficient of lift slope, radians  
 $C_L$  – Three-dimensional coefficient of lift, dimensionless  
 $C_{L,\alpha}$  – Three-dimensional theoretical coefficient of lift slope, radians  
 $C_p$  – Coefficient of pressure, dimensionless  
 $h$  – horn height, mm  
 $K_0$  – Modified inertia parameter, dimensionless  
 $L$  – Length proportional to model chord, cm  
 $l$  – Characteristic length, m  
 $LWC$  – Liquid water content of cloud, g/m<sup>3</sup>  
 $MVD$  – Water drop median volume diameter,  $\mu\text{m}$   
 $n_o$  – Stagnation freezing fraction, dimensionless  
 $s$  – Horn location surface length, mm  
 $s/c$  – Non-dimensional surface length  
 $t$  – Temperature, °C  
 $We_\delta$  – Weber number based on drop size and water properties, dimensionless

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$We_L$  – Weber number based on length,  $L$ , and water properties, dimensionless  
 $\alpha$  – Angle of Attack, degrees  
 $\beta_0$  – Stagnation collection efficiency, dimensionless  
 $\Phi$  – Drop energy transfer parameter, °C  
 $\Theta$  – Horn angle with respect to chord line, degrees  
 $\Theta_{ice}$  – Air energy transfer parameter, R

$\tau$  – Accretion time, min

#### General Subscripts:

$0$  – Stagnation point

$st$  – Static

$tot$  – Total

## II. Introduction

Aircraft icing is a crucial area of study to ensure operational safety. Due to the safety concern icing poses, all Federal Aviation Administration (FAA) certified aircraft must be operable under the icing conditions specified by Title 14 of the FAA Code of Federal Regulations (CFR), Part 25, Appendix C envelope [1] and the recently added Appendix O envelope [1], referred to as the Appendix C envelope and the Appendix O envelope, respectively in this paper. Multiple tools are utilized to study aircraft icing, including predictive software, such as LEWICE, and icing research tunnels. LEWICE [2] is a predictive software developed by the NASA Glenn Research Center (GRC) to predict the ice formation on aircraft surfaces. Due to the geometric limitation of icing research tunnels and the required icing studies performed prior to flight tests, a recommended scaling method by Anderson and Tsao [3] is utilized to adapt icing tunnel parameters (such as the flow temperature, the test section flow velocity, the water droplet median volume diameter (MVD), the liquid water content (LWC), and the ice accretion time,  $\tau$ , etc.), along with model scaling to ensure the ice accretion collected in the wind tunnel is representative of the full-scale ice formation. The objective of both these methods is to produce an ice accretion that provides dynamic similitude of the ice accretion formed on a full-scale model or aircraft, which is crucial in recreating the aerodynamics and aerodynamic penalties associated with icing. Throughout the paper, the ice accretion collected on a full-scale wing at NASA Glenn is referenced as the “full-scale ice shape”, the ice accretion collected on a scaled model using the recommended ice shape scaling method by Anderson and Tsao [3] is referred to as the “scaled ice shape”, and the predicted ice accretion by the LEWICE software [2] is referred to as the “LEWICE ice shape.” The full-scale ice accretion refers to an ice accretion collected on a full-scale model, a model with a 1:1 representation of a desired body of interest.

### A. Ice Formations in Appendix C and SLD Conditions

The ice formations in Appendix C are typically broken into two primary regions, the main and the feather region ice formations. As mentioned by Anderson [4], the primary objective of the ice shape scaling method is to preserve the main region of the leading-edge ice shape as it is the most dominant feature geometrically and has the greatest influence on an airfoil’s aerodynamics. Similarly, the LEWICE software only predicts the main ice formation due to its prominence and due to the flow dynamics in the feather region not being well understood. Though the main region of ice accretions is the most dominant and influential in Appendix C envelope conditions, Anderson and Tsao [3], showed that the large MVD of SLD conditions generate large feather region icing compared to Appendix C conditions, the effects of which cannot be neglected. The differences in the ice shapes formed under Appendix C and SLD conditions is shown in Figure 1 for a full-scale and a scaled down model.

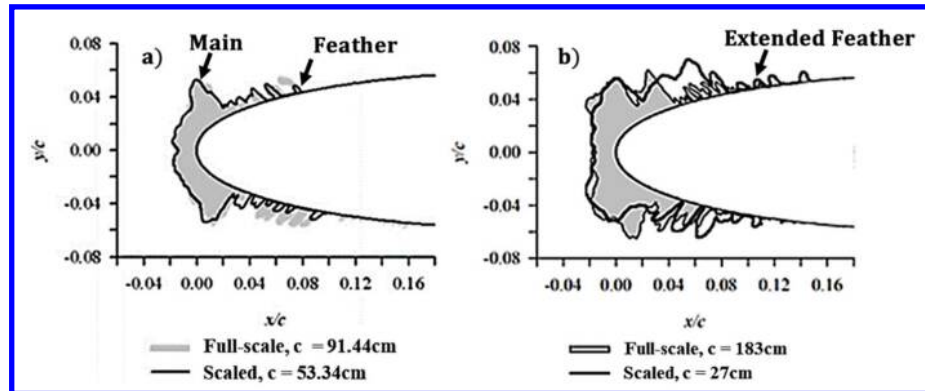


Figure 1 Scaled and full-scale ice accretion used in the verification for a) Appendix C conditions [4] and b) for SLD conditions [3]

The ice shapes in Figure 1 are tracings collected at the midspan of the NACA 0012 models in the NASA Glenn IRT. This method does not capture the spanwise variation of the ice shape, primarily in the feather region, or the surface roughness. Though these are simplified representations of the ice accretion, they were consistently collected at the midspan and still represent the geometric variation between the full-scale and scaled ice shapes.

Figure 1 shows that the scaling method is highly effective in reproducing the ice shape under Appendix C conditions, but not to the same degree under SLD conditions. Anderson and Tsao [3] attributed this to the larger feather region accretions that occur under SLD conditions due to the large MVD. Both the scaled and full-scale ice shapes in Figure 1b, have large feather region ice accretions, especially on the upper surface of the scaled ice shape.

Conditions with large MVD or high LWC, as found in Appendix O SLD conditions, commonly result in horn ice geometries. Horn ice geometries can feature either a single or a double horn, such as those found in Figure 1. Double horn ice accretions are typically generated at lower, more neutral angles of attack. The schematic in Figure 2, provided by Bragg et al. [5], outlines the geometric parameters used to characterize a horn ice accretion. The horn shape is characterized by three geometric parameters: the angle it makes with respect to the chord line,  $\theta$ , the horn height,  $h$ , and the non-dimensional surface length,  $s/c$ , where  $s$  is the surface length location of the horn shape and  $c$  is the chord length [5]. These parameters all have a large effect on the aerodynamic penalties associated with horn ice accretions. The horn ice geometry is strongly influential on the airfoil aerodynamics, generating primarily two-dimensional flow separation.

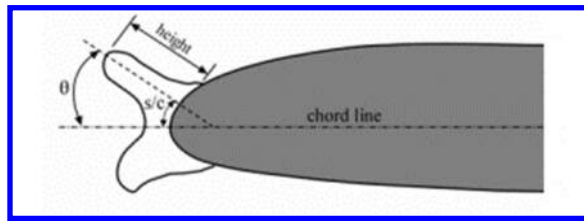


Figure 2 Horn Ice Geometry Schematic [5]

## B. Recommended Ice Shape Scaling Method

Due to test section geometric constraints and icing limitations, scaling methods to scale both the model size and icing test parameters are employed to study the formation and accretion of ice on aircraft surfaces, especially on the wing leading edge. These methods are designed to ensure the ice shape produced, specifically the main leading-edge ice formation, is non-dimensionally characteristic of the ice accretion formed on a full-scale model. Scaling methods developed by Ruff [6] and Olsen [7] were evaluated in *Manual of Scaling Methods* by Anderson [4]. The scaling method determined from Anderson's evaluation [4] of methods proposed by Ruff [6] and Olsen [7], has since been revised and updated. The current recommended ice shape scaling method was evaluated and determined by Anderson and Tsao [3] at the NASA Glenn IRT. It is based on the initial method provided by Anderson [4], but allows for more flexibility in scaling the physical test conditions.

The recommended ice shape scaling method utilizes a modified version of the method proposed by Ruff [6], in which the modified inertia parameter,  $K_0$ , the accumulation parameter,  $A_c$ , the stagnation freezing fraction,  $n_0$ , and the Weber number based on the model length and the water properties,  $We_L$ , are matched between the scale and full-scale (reference). Through the similarity parameters, the wind tunnel conditions required to generate a characteristic ice shape are determined. The wind tunnel flow parameter determined include the temperature of the flow, the test section freestream velocity, MVD, LWC, and the ice accretion time,  $\tau$ .

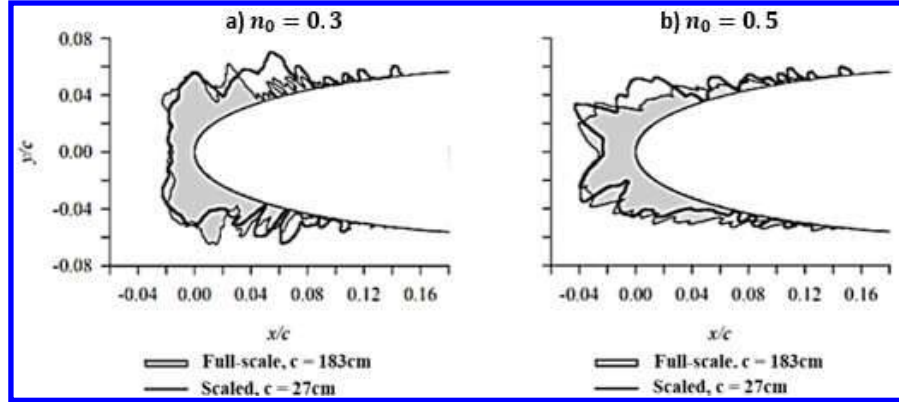
## C. LEWICE

The LEWICE software is an ice accretion predictive software based on an analytical model of the thermodynamics involved in icing. Since the flow dynamics in the feather region is not well understood, LEWICE only focuses on the prediction of frontal ice shape. The model utilizes a significant number of atmospheric and IRT conditions to predict the ice shape formation on numerous aircraft surfaces and can even include the effects of different deicing methods. The goal of the software is to accurately predict the location, size, and shape of an ice accretion based on surface geometry and icing conditions [2]. LEWICE provides a method for studying multiple aircraft icing situations without the necessity for extensive experimentation in an IRT and can predict two-dimensional or three-dimensional ice accretions.

## D. Research Objectives

The objective of this research is to provide an initial low Reynolds number aerodynamic verification the ice accretions generated by the LEWICE software and ice shape scaling method under SLD conditions by comparing the

surface and near wake aerodynamics with that of the full-scale ice accretion under two different stagnation freezing fractions,  $n_0$  of 0.3 and 0.5. The stagnation freezing fraction,  $n_0$ , is the ratio of the amount of water that freezes versus the amount of water that reaches the stagnation point.  $n_0$  has a large influence on the horn geometry of the ice accretion with higher  $n_0$  generating ice shapes with lower horn angles and more acute s/c positioning. The ice accretions at  $n_0$  of 0.3 and  $n_0$  of 0.5 are shown in Figure 3.

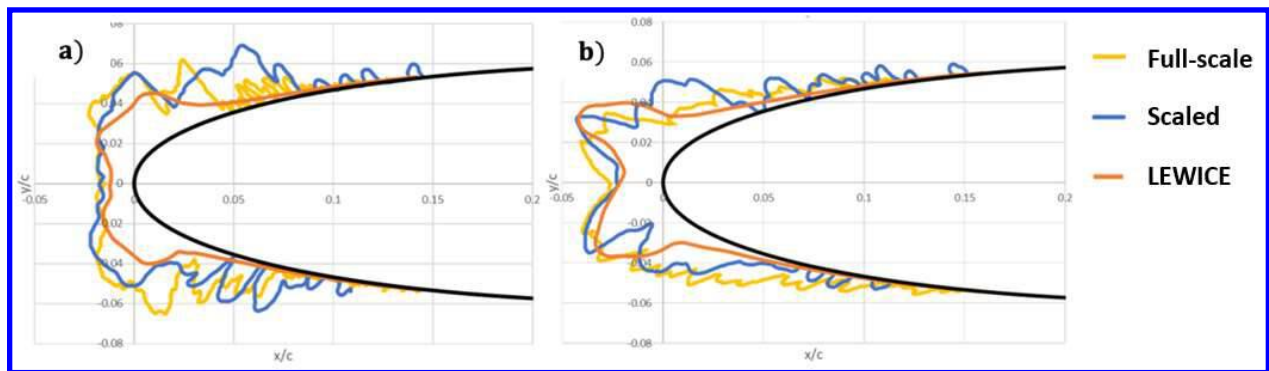


**Figure 3 Full-scale and scaled non-dimensional ice shapes used for scaling method verification with a) Ice shapes collected at  $n_0 = 0.3$  and b) Ice shapes collected at  $n_0 = 0.5$  [3]**

The ice shapes in Figure 3 were chosen based on ice shape verification trials from Anderson and Tsao [3], with the specified icing conditions outlined in Table 1. These conditions, along with the droplet diameter distribution based on total LWC supplied by the NASA GRC IRT, were used to generate the corresponding LEWICE ice structures. The full-scale ice shape refers to the ice accretion collected on the 183cm chord NACA 0012 airfoil and the scaled ice shape refers to the ice accretion collected on the 27cm chord NACA 0012 airfoil in this paper. The input files used to generate the LEWICE predicted accretions can be found in Appendix A. The droplet diameter distribution supplied by NASA GRC IRT for an MVD of 170 $\mu$ m can be found in Appendix B. Overlays of the corresponding ice accretions are graphed in Figure 4.

**Table 1 Icing Conditions for Ice Shapes in Figure 2 [3]**

| Runs                              | $C$<br>[cm] | $t_{st}$<br>[°C] | $t_{tot}$<br>[°C] | $V$<br>[kt] | $MVD$<br>[ $\mu$ m] | $LWC$<br>[g/m] | $\tau$<br>[min] | $\beta_0$<br>[%] | $A_c$ | $\beta_0 A_c$ | $n_0$ | $We_{\delta}$ ,<br>$10^3$ | $We_L$ ,<br>$10^6$ |
|-----------------------------------|-------------|------------------|-------------------|-------------|---------------------|----------------|-----------------|------------------|-------|---------------|-------|---------------------------|--------------------|
| <b>(a) <math>n_0 = 0.3</math></b> |             |                  |                   |             |                     |                |                 |                  |       |               |       |                           |                    |
| 1                                 | 182.9       | -12              | -11               | 100         | 170                 | 1.45           | 22.9            | 92.3             | 1.93  | 1.78          | 0.30  | 6.90                      | 2.35               |
| 2                                 | 26.7        | -12              | -2                | 268         | 33                  | 0.86           | 2.1             | 92.3             | 1.94  | 1.79          | 0.30  | 9.59                      | 2.47               |
| <b>(b) <math>n_0 = 0.5</math></b> |             |                  |                   |             |                     |                |                 |                  |       |               |       |                           |                    |
| 1                                 | 182.9       | -21              | -21               | 100         | 170                 | 1.45           | 22.9            | 92.2             | 1.93  | 1.78          | 0.50  | 6.90                      | 2.35               |
| 2                                 | 26.7        | -15              | -5                | 276         | 31                  | 0.64           | 2.7             | 92.0             | 1.89  | 1.74          | 0.50  | 9.71                      | 2.61               |



**Figure 4 Ice shape overlay of ice accretions collected with a)  $n_0 = 0.3$  and b)  $n_0 = 0.5$**

The flow field investigations discussed in this paper addresses the aerodynamic similarities and differences between the ice accretions shown in Figure 4.

### III. Experimental Setup

#### A. University of Dayton Low Speed Wind Tunnel (UD-LSWT)

All experimentation was performed in the University of Dayton Low Speed Wind Tunnel (UD-LSWT). The UD-LSWT has a 16:1 contraction ratio, 6 anti-turbulence screens, and can be operated in five different configurations: four different interchangeable test sections (76.2cm x 76.2cm x 243.8cm) and an open jet configuration. In all five configurations, the tunnel can obtain a free stream velocity range of 6.7 m/s to 40m/s with a turbulence intensity below 0.1% at 15m/s measured using a hot-wire anemometer. The free stream velocity profile can be varied up to 5Hz and over 50% velocity amplitude using a downstream shuttering system. All experiments were performed in the open jet configuration, where the inlet opens into a sealed plenum. The effective length in this configuration is 182cm. The expanded flow is collected by a 137cm x 137cm collector prior to a diffuser section and the drive fan. The velocity variation for a given RPM of the drive fan is found using a Pitot tube connected to an Omega differential pressure transducer (Range: 0-6.9kPa).

#### B. Horn Ice Models

Clear geometric differences are visible between the different ice shapes shown in Figure 4. Table 2 summarizes the main region ice geometry of the different ice accretions of interest. They are separated into two sets based on their associated stagnation freezing fraction. The geometry of the horn ice accretion varies significantly with  $n_0$ . As such, it was chosen as a parameter of interest in the present investigation to perform aerodynamic verification. It is important to note that the ice shapes are not symmetric. The values in Table 2 were calculated based on a point 0.01c from the leading edge along the chord. As mentioned, the stagnation freezing fraction has a large influence on the s/c and horn angle of the horn ice shape geometry.

*Table 2 Summarization of Ice Shape Geometries Tested*

|                               | Horn Designation | $\Theta$ , [deg] | s/c    | h/c   |
|-------------------------------|------------------|------------------|--------|-------|
| Case 1: Baseline NACA 0012    | N/A              | N/A              | N/A    | N/A   |
| <b><math>n_0 = 0.3</math></b> |                  |                  |        |       |
| Case 2: Full-scale            | Upper            | 80.7             | 0.017  | 0.041 |
|                               | Lower            | 89.4             | 0.020  | 0.047 |
| Case 3: Scaled                | Upper            | 79.5             | 0.017  | 0.042 |
|                               | Lower            | 74.1             | 0.015  | 0.039 |
| Case 4: LEWICE                | Upper            | 82.3             | 0.018  | 0.029 |
|                               | Lower            | 83.4             | 0.018  | 0.024 |
| <b><math>n_0 = 0.5</math></b> |                  |                  |        |       |
| Case 5: Full-scale            | Upper            | 31.1             | 0.0056 | 0.042 |
|                               | Lower            | 35.1             | 0.0063 | 0.045 |
| Case 6: Scaled                | Upper            | 31.8             | 0.0057 | 0.051 |
|                               | Lower            | 35.4             | 0.0064 | 0.048 |
| Case 7: LEWICE                | Upper            | 34.9             | 0.0063 | 0.051 |
|                               | Lower            | 39.8             | 0.0073 | 0.043 |

A model for each case was designed in SolidWorks and 3D printed using Nylon SLS. The models have a span and chord length of 0.2032m, providing a model AR of 1. This chord length was chosen to allow for the preservation of the fine ice features. Two-dimensional flow was achieved by adding end plates to the wingtips. The experimental setup will be discussed in the following sections.

#### C. Particle Image Velocimetry Overview

The seeding in the UD-LSWT was performed using a Vicount smoke seeder with glycerin-based oil, resulting in tracer particles with a mean diameter of 0.2 microns. The Stokes number of the glycerin-based oil was calculated based on the experimental parameters. Sea level conditions were assumed for the density and dynamic viscosity, 1.225 kg/m<sup>3</sup> and 1.81e-5 kg/(m•s), respectfully. The freestream velocity of all PIV experiments was 22.06m/s, based on a



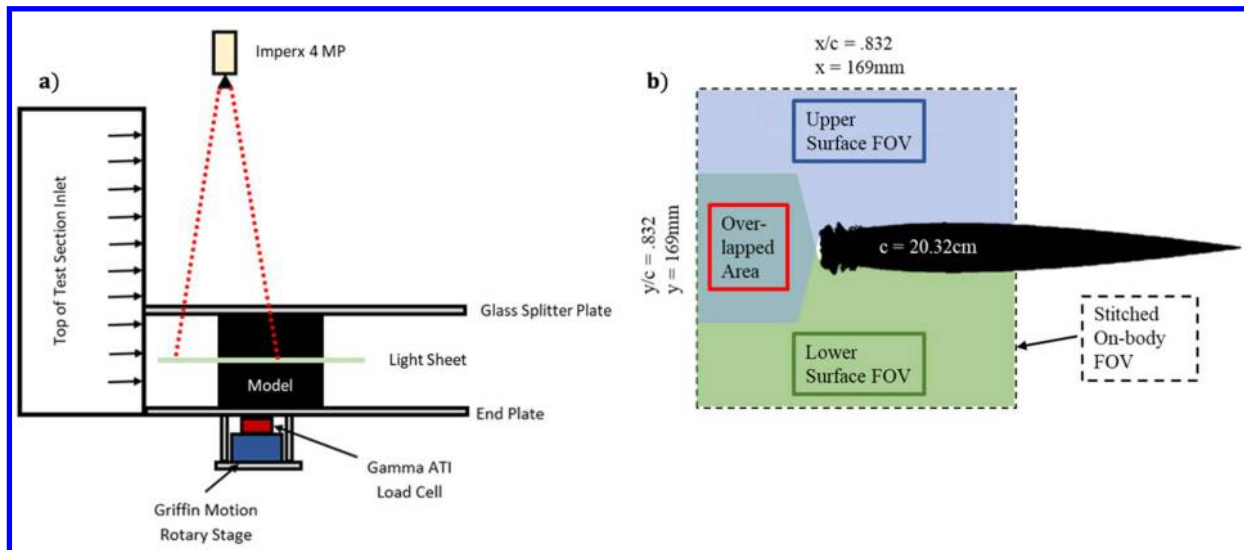
Reynolds number of 300,000. The thickness of the models, based on the NACA 0012 profile used, was .0245 m. This results in a Stokes number of  $1.35e-7$ , which is significantly less than 1, as desired. Nd: YAG frequency doubled laser (Quantel Twins CFR 300) was used as a light source. The Imperx 4 MP high speed camera was used to capture the particle images. In each test case, 1500 image pairs were obtained for the free-shear PIV and 1000 image pairs were obtained and processed for the on-body PIV. The uncertainty in the velocity measurements determined through the PIV image capturing and image processing through DPIV was determined using the process outlined by Lazar et al. [8]. Using this process and the parameters used in the PIV setup, the overall mean uncertainty was determined to be 1.53% in the flow field velocities, approximately  $\pm 0.338\text{m/s}$  in the freestream flow.

#### D. On-Body PIV Setup in the UD-LSWT

On-body PIV was completed to study the flow field near and around the ice shape allowing for assessment of the ice shape effect on the local aerodynamics. This allowed for studying separation bubble size, flow separation, reattachment, and shedding. It also allowed for the assessment of secondary separation points, if present, due to the large feather region ice accretions commonly found in high MVD conditions. The test matrix for the on-body PIV is presented in Table 3. Figure 5 shows a schematic of the on-body setup in the UD-LSWT, along with the on-body field of view (FOV). In order to simulate two-dimensional flow, as best as possible, an optics glass plane was used as the splitter plate between the camera and model. The glass splitter was able to be mounted directly against the end surface of the model.

**Table 3 On-body PIV Test Matrix**

| On-Body         |                            |           |                                |                      |                      |
|-----------------|----------------------------|-----------|--------------------------------|----------------------|----------------------|
| Reynolds Number | Freestream Velocity, [m/s] | Cases     | AOA Sweep                      | Number of Iterations | Image Pairs Captured |
| 300,000         | 22.06                      | All Cases | -12° to 12°<br>(3° increments) | 2                    | 1000                 |



**Figure 5 a) Schematic of on-body PIV setup in UD-LSWT b) On-body FOV position with respect to the model schematic. The upper and lower surface PIV was performed separately.**

The on-body PIV parameters are presented in Table 4. The magnification factor was calculated to be 13.2. The  $\Delta t$  was optimized based on achieving a 10 to 11 particle pixel displacement in the freestream flow, while maintaining a minimum of 4 to 5 particle pixel displacement in the separation bubble. The laser sheet opened from beneath the model. Due to the placement of the laser sheet, image collection could only be performed on one surface of the model at a time. To collect PIV images on the other surface, the whole setup was flipped around the centerline of the tunnel, while keeping the PIV parameters the same. The processed upper and lower image pairs were then stitched together to study the complete near ice accretion flow field. The images were stitched by finding a region of overlapping velocity vectors, see Figure 5, between the corresponding processed upper and lower surface PIV data,

using MATLAB. The streamwise velocity, Reynolds stress, and the on body coherent structure contours were then overlapped and stitched based on the positioning determined through evaluating the velocity vectors.

**Table 4 On-body PIV Parameters**

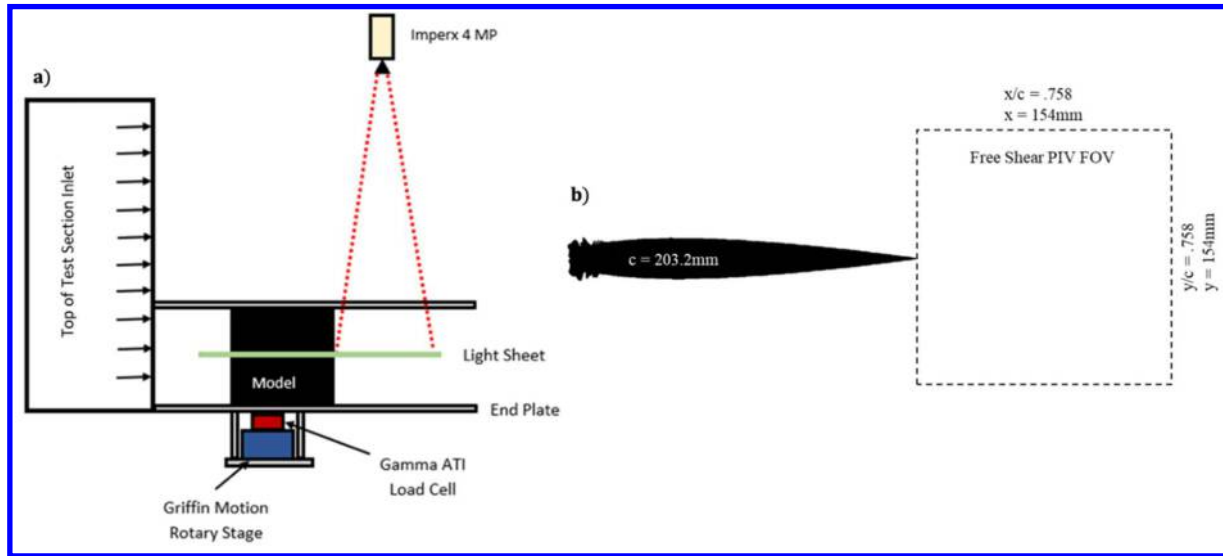
| On-Body                               |               |                       |                                      |                          |                 |           |        |
|---------------------------------------|---------------|-----------------------|--------------------------------------|--------------------------|-----------------|-----------|--------|
| Imperx 4 MP CCD dim., [pixel x pixel] | FOV [mm x mm] | FOV in terms of chord | Spatial Resolution of FOV, [pixel/m] | $\Delta t$ , [ $\mu s$ ] | $\lambda$ , [m] | $f$ , [m] | $\psi$ |
| 2048 x 2048                           | 169 x 169     | .832c x .832c         | 12118                                | 36                       | 1.32            | 0.1       | 13.2   |

## F. Free Shear Layer PIV setup in the UD-LSWT

The near wake free shear layer was studied to assess the effect of ice accretions on the airfoil's flow field. The near wake flow characteristics are strongly coupled with the dynamics of the upper and lower surface boundary layers and can be used to determine if the scale and LEWICE accretions faithfully simulate the corresponding upper and lower surface boundary layer characteristics of the full-scale ice accretion. Studying the near wake location allowed for the size of the wake to be assessed, along with the Reynolds stress and the coherent structures present in the wake. The test matrix for the free shear layer PIV is presented in Table 5. A schematic of the on-body setup in the UD-LSWT is shown in Figure 6, along with the associated FOV. In order to simulate two-dimensional flow, as best as possible, a thin acrylic sheet with a refraction index similar to optical glass was used as the splitter plate between the camera and model. The acrylic splitter was able to be mounted directly against the end surface of the model.

**Table 5 Free Shear Layer PIV Test Matrix**

| Near Wake Free Shear Layer |                            |           |                             |                      |                      |
|----------------------------|----------------------------|-----------|-----------------------------|----------------------|----------------------|
| Reynolds Number            | Freestream Velocity, [m/s] | Cases     | AOA Sweep                   | Number of Iterations | Image Pairs Captured |
| 300,000                    | 22.06                      | All Cases | -14° to 14° (2° increments) | 2                    | 1500                 |



**Figure 6 a) Schematic of free shear layer setup in UD-LSWT b) Free Shear Layer FOV position with respect to the model schematic**

The free shear PIV parameters are presented in Table 6. The same 105mm lens was used in collecting the free shear layer PIV images as for the on-body PIV image capture. The  $\Delta t$  was optimized based on achieving a 10 to 11 particle pixel displacement in the freestream flow, while maintaining a minimum of 4 to 5 particle pixel turbulent

region of the wake. The position and size of the FOV allowed for the wake to be completely captured with the freestream velocity still in view.

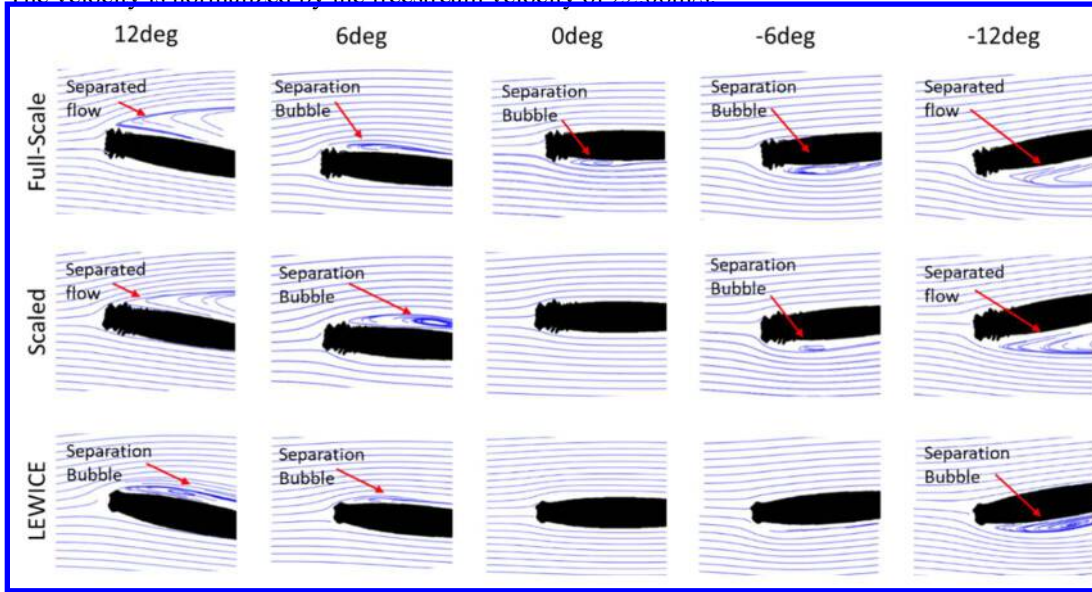
**Table 6 Free Shear Layer PIV Parameters**

| Near Wake Free Shear Layer               |                  |                          |                                          |                             |                    |              |        |
|------------------------------------------|------------------|--------------------------|------------------------------------------|-----------------------------|--------------------|--------------|--------|
| Imperx 4 MP CCD<br>dim., [pixel x pixel] | FOV<br>[mm x mm] | FOV in terms of<br>chord | Spatial Resolution of<br>FOV, [pixels/m] | $\Delta t$ ,<br>[ $\mu s$ ] | $\lambda$ ,<br>[m] | $f$ ,<br>[m] | $\psi$ |
| 2048 x 2048                              | 155 x 155        | .758c x .758c            | 13213                                    | 40                          | 1.21               | 0.1          | 12.1   |

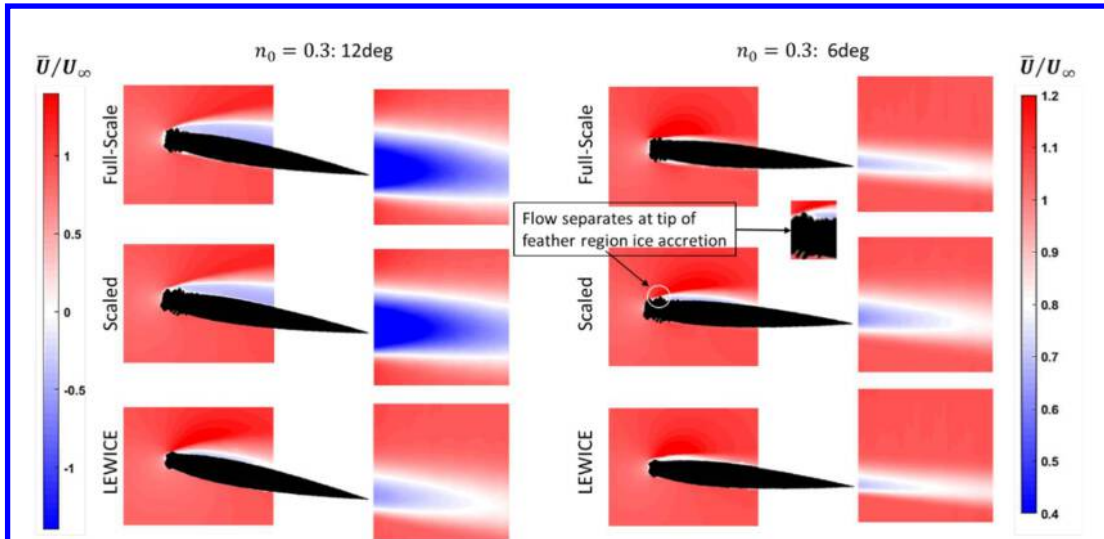
## V. PIV Flow Field Analysis

### A. Mean U-velocity Component: $n_0 = 0.3$ Ice Accretions

The on-body streamlines for each of the ice shapes are shown in Figure 7, where the presence of a separation bubble and complete flow separation can be observed. The normalized U-component velocity contours for the ice accretions collected at a  $n_0 = 0.3$  is shown in Figure 8. It is important to note that the contour map range is different between the on-body and free shear layer contours. The contour range for the on-body contours is provided on the left-hand side of the figure and the contour range for the free shear layer is provided on the right-hand side of the figure. The velocity is normalized by the freestream velocity of 22.06m/s.



**Figure 7 On-body streamlines for  $n_0 = 0.3$  ice shapes**





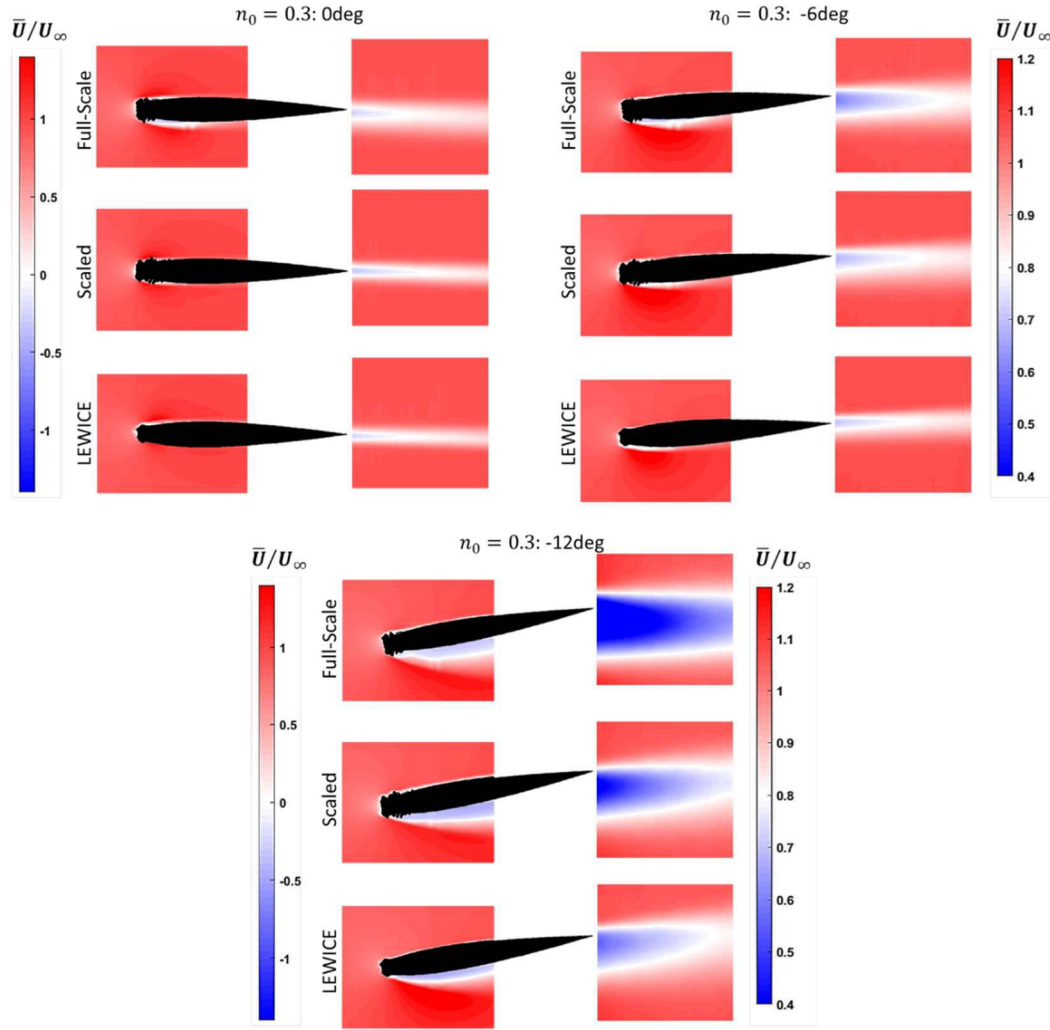


Figure 8 Normalized  $U$ -velocity contours for  $n_0 = 0.3$  ice shapes

At the positive  $12^\circ$  case, the flow is separated for the full-scale and scaled ice accretions, but not the LEWICE ice shape, as annotated in Figure 7. A clear separation bubble is seen on the upper surface of the LEWICE ice shape, annotated in Figure 7, and is responsible for the flow remaining attached at  $12^\circ$ . The near wake flow physics show a thicker wake width in the case of the full-scale and scale ice accretion compared to the LEWICE ice accretion. This is associated with a larger increase in drag in the full-scale and scaled model when compared to the LEWICE model.

At a positive  $6^\circ$  angle of attack, the scaled and full-scale ice shapes deviate from one another. Though they both show a separation bubble on the upper surface, the separation bubble is significantly larger for the scaled ice shape. This causes flow reattachment to occur significantly further aft on the scaled ice shape airfoil causing a thicker boundary layer. This difference between the full-scale and scaled ice shapes is also shown by the larger wake of the scaled ice shape. The scaled and full-scale upper horns are of similar geometry, suggesting the large feather region structure present on the upper surface of the scaled ice accretion is affecting the flow field. As annotated in Figure 8, this was shown to be the case, as separation occurs at the tip of the feather region ice accretion. The LEWICE ice shape creates an upper surface separation bubble similar in size to the full-scale ice shape, but neither the scaled or LEWICE ice shapes show separation on the lower surface. The full-scale ice shape, on the other hand, does have a small region of separation on the lower surface, resulting in a slightly larger wake than the LEWICE ice shape. Just like at  $6^\circ$ , the full-scale ice shape has a larger separation bubble on the lower surface than either the scaled or LEWICE structures at  $0^\circ$ .

The larger separation bubble is caused by the taller lower horn of the full-scale ice shape. Similar to the  $0^\circ$  case, the larger lower surface separation bubble continues to persist at  $-6^\circ$  angle of attack with respect to the full-scale

ice shape. This results in a slightly thicker wake width, as shown in the wake contours, when compared to the scaled and LEWICE ice shapes. Though the flow on both the full-scale and scaled ice shapes is fully separated at  $-12^\circ$ , the momentum lost in the wake is stronger in the full-scale ice shape. Flow on the lower surface of the LEWICE ice shape has not yet separated, as shown by the streamlines in Figure 7. The large separation bubble results in significantly less momentum loss in the wake.

### B. Mean U-velocity Component: $n_0 = 0.5$ Ice Accretions

The on-body streamlines for each of the ice shapes collected at a  $n_0 = 0.5$  are shown in Figure 9. The streamlines allowed the presence of a separation bubble versus complete flow separation to be determined. Figure 10 includes the normalized U-component velocity contours for the ice accretions collected at a  $n_0 = 0.5$ . It is once again important to note that the contour map range differed between the on-body and free shear layer contours. The contour range for the on-body contours is provided on the left-hand side of the figure and the contour range for the free shear layer is provided on the right-hand side of the figure. The velocity is normalized by the freestream velocity of 22.06m/s

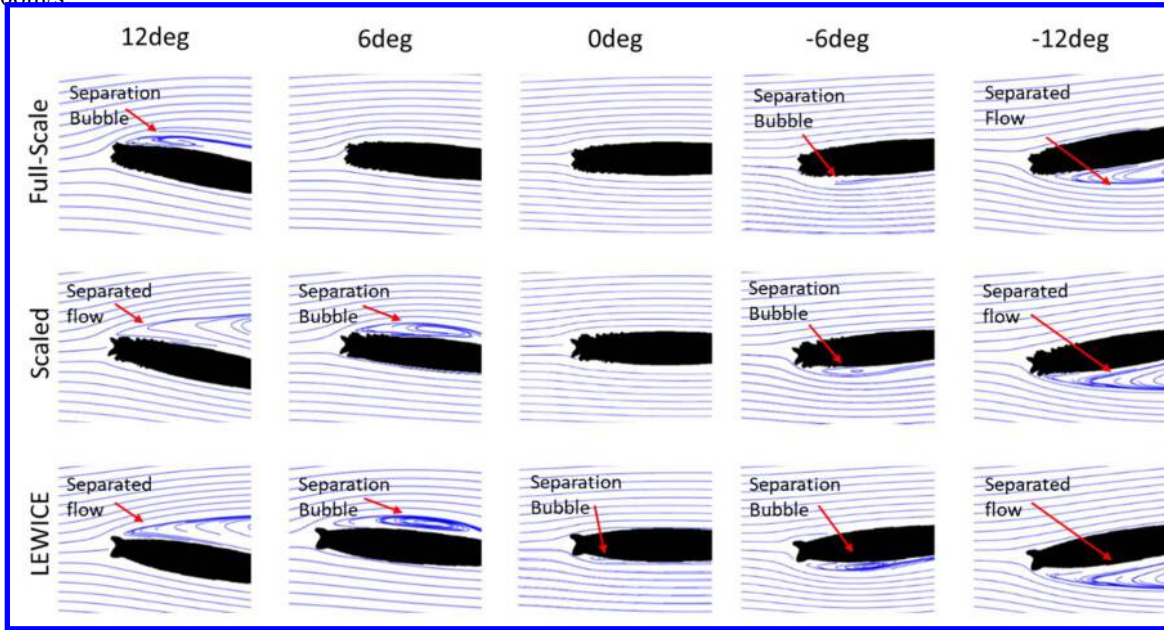
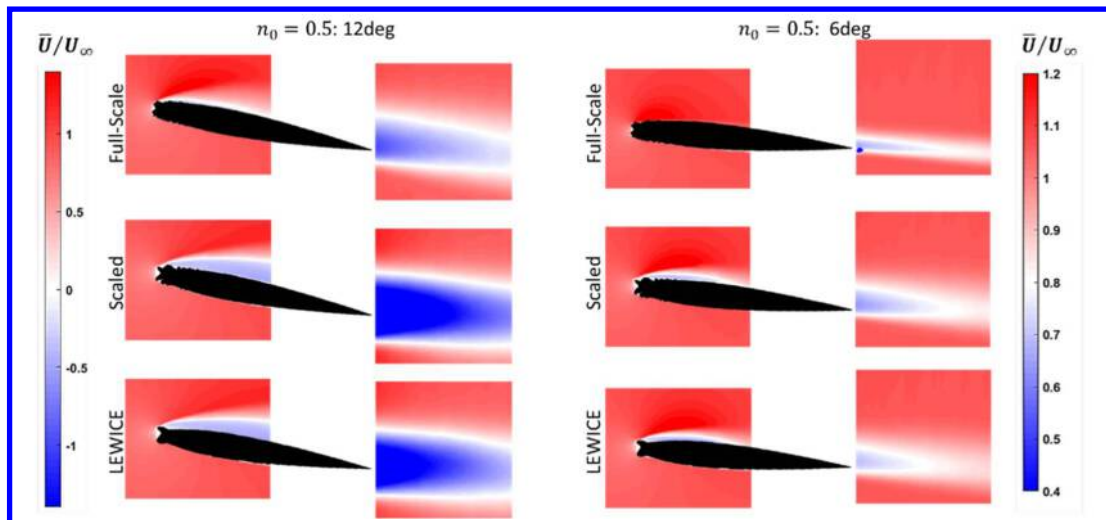


Figure 9 On-body streamlines for  $n_0 = 0.5$  ice shapes



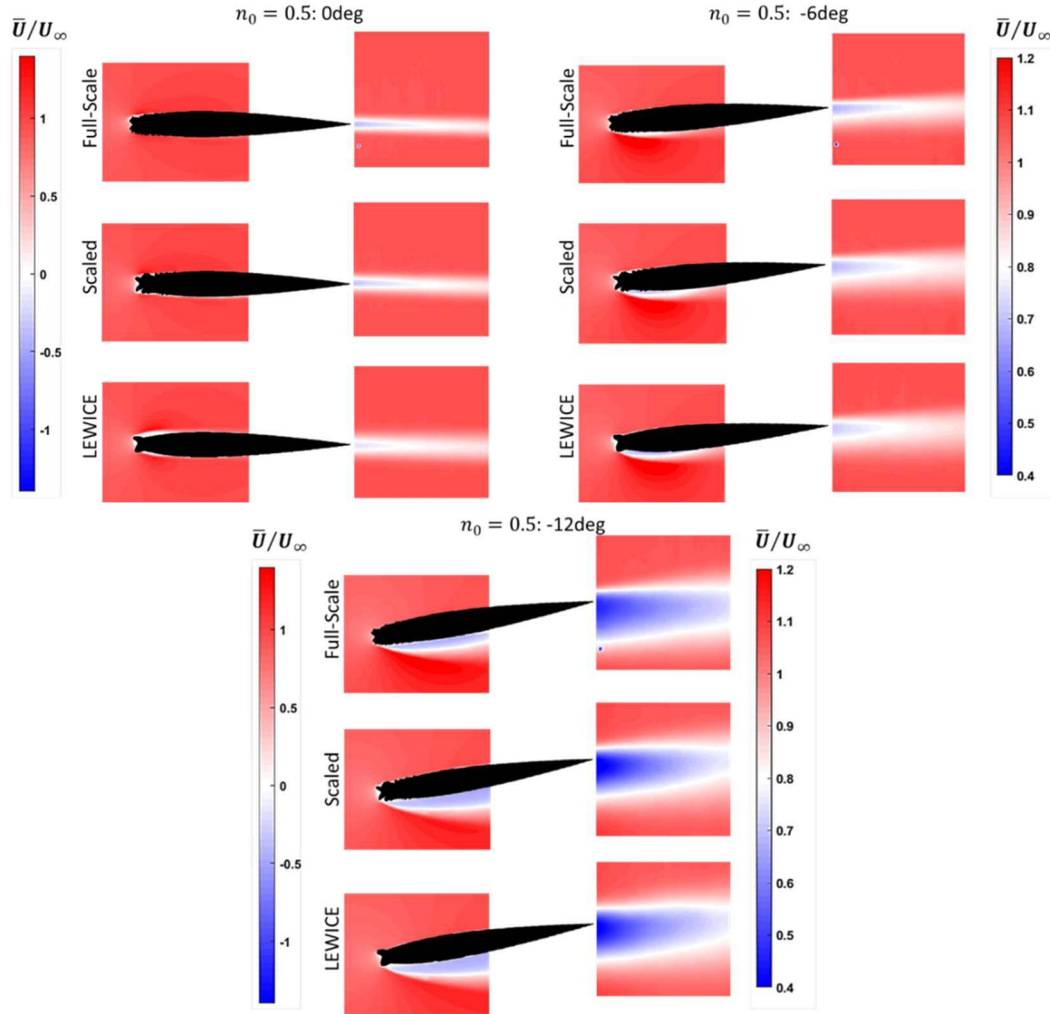


Figure 10 Normalized U-velocity contours for  $n_0 = 0.5$  ice shapes

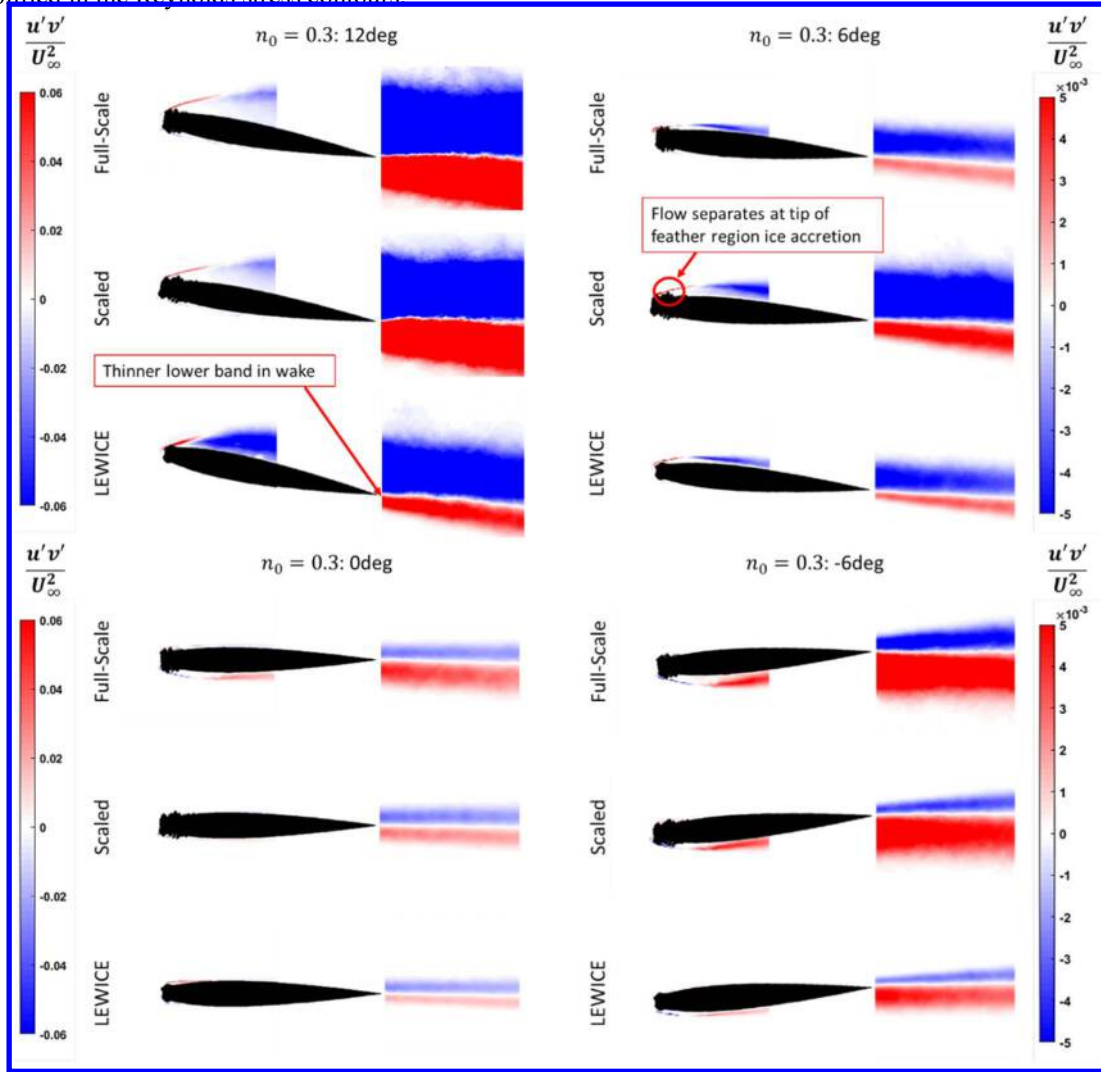
At an angle of attack of  $12^\circ$ , the flow is fully separated for the LEWICE and scaled ice shapes, but not the full-scale, as annotated in Figure 9. A separation bubble on the upper surface of the full-scale can be seen in Figure 9, which could have reattached near the trailing edge preventing complete flow separation. The near wake flow physics show thicker wake widths with respect to the scaled and LEWICE ice shapes than the full-scale. At  $6^\circ$ , the scaled and LEWICE ice shapes both have a similar large upper surface separation bubble, whereas no visible separation is present on the upper surface of the full-scale ice shape. The lack of separation causes the wake width of the full-scale ice shape to be smaller than the other two ice shapes. The LEWICE ice shape is the only ice shape that shows a noticeable separation bubble at  $0^\circ$  angle of attack, as seen in Figure 9. This results in a wake width thicker than the full-scale and scaled ice shapes. All three ice shapes show a separation bubble on the lower surface at  $-6^\circ$ , as annotated in Figure 9. The near wake flow physics show wakes of similar widths and strengths, with the full-scale wake width being slightly thinner than the scaled and LEWICE ice shapes. The wake physics of the three ice shapes are again similar at  $-12^\circ$ , with the LEWICE wake being slightly stronger than the other two ice shapes. On-body, all three ice shapes show flow separation, but the full-scale separation region is significantly shorter than separation caused by the LEWICE and scaled ice shapes.

### C. Reynolds Stress: $n_0 = 0.3$ Ice Accretions

The flow in the upper and lower surface of wings with ice accretions as well as the near wake is inherently turbulent in nature. As such, it is prudent to analyze both the mean and fluctuating component of the flow field. In order to compare the turbulent flow properties, a comparison of the Reynolds shear stress of the ice shape flow fields is presented in the same style as the mean U-velocity. The Reynolds shear stress represents the momentum transport due to turbulence generated by the ice accretions. The magnitude of the Reynolds shear stress represents the correlation

between the streamwise fluctuating velocity component ( $u'$ ) and the wall normal fluctuating velocity component ( $v'$ ). A higher magnitude Reynolds stress represents a higher correlation between these two quantities. The sign of the Reynolds stress represents the direction of the correlation. Positive Reynolds stress indicates both fluctuating velocities have positive values, whereas the negative Reynolds stress indicates either one of the fluctuating velocities have a negative sign. In the present context, the magnitude of the Reynolds stress is more significant than its direction. Since the Reynolds stress represents the momentum transport due to turbulence, any minor deviations in the flow field due to the different ice accretions are amplified. The Reynolds stress discussed in this section is normalized by the square of freestream velocity.

Figure 11 includes the Reynolds stress contours for the ice accretions collected at  $n_0 = 0.3$ . It is important to note that the contour range differs between the on-body and free shear layer contours. The contour range for the on-body contours is provided on the left-hand side of Figure 11 and the contour range for the free shear layer contour is provided on the right-hand side of Figure 11. The differences seen between the three cases in the normalized U velocity is amplified in the Reynolds stress contours.





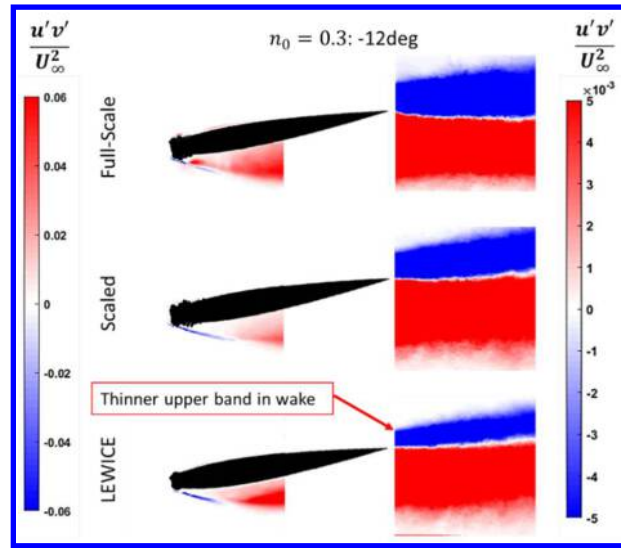


Figure 11 Reynolds stress contours for  $n_0 = 0.3$  ice shapes

Strong negative Reynolds stress is observed in the upper surface of the LEWICE shape when compared to the full-scale and the scaled ice shapes at  $12^\circ$ . It is interesting to note that the higher magnitude of the Reynolds stress is seen closer to the leading edge in the LEWICE shape when compared to the other two cases. A strong Reynolds stress could also indicate the presence of smaller coherent length scales when compared to the other two ice shapes, which will be confirmed by performing modal analysis. The Reynolds stress bands in the near wake also show similar trends between the full-scale and the scaled wing model when compared to the LEWICE. It should be noted that the wake Reynolds stress corresponding to the lower surface of the wing (red region) has similar thickness between the full-scale and the scaled ice shape, whereas the thickness is significantly thinner in the LEWICE shape. This shows less momentum transport due to turbulence from the flow exposed surface of the LEWICE ice shape compared to the other two ice shapes.

As seen in the U-velocity contours, similar deviations are observed in the Reynolds stress between the full-scale and scaled ice shapes at  $6^\circ$  angle of attack. A clear separation point at tip of large feather accretion is seen on the scaled ice shape. At  $6^\circ$  angle of attack, the LEWICE ice shape is characteristic of the full-scale, having similar turbulence transition points in the shear layer generated by the upper surface ice shape. The Reynolds stress in the wake is highly similar between the LEWICE and full-scale, whereas the Reynolds stress in the wake of the scaled ice shape is significantly greater.

The  $0^\circ$  angle of attack mean velocity contours showed that the LEWICE shape is more streamlined than the full-scale and the scaled models. The presence of a larger lower separation bubble in the lower surface of the full-scale model is not successfully replicated by either the scaled or the LEWICE ice shapes. This led to the differences seen in the near wake where the full-scale ice shape Reynolds stress corresponding to the lower surface (red band) has a thicker width when compared to the lower surface Reynolds stress from the other two cases. The magnitude of the Reynolds stress from the LEWICE shape is much lower than that of the full scale.

The near wake Reynolds stress thickness and magnitude associated with the lower surface between the full-scale and scaled ice shapes compare extremely well at  $6^\circ$  angle of attack with the LEWICE ice shape failing to match this trend. The LEWICE ice shape produces significantly less turbulence in the wake and on-body than the other two ice shapes.

The near wake Reynolds stress thickness and magnitude associated with the lower surface compares well for all three ice shapes at  $-12^\circ$ . The Reynolds stress band in the near wake due to the upper surface is noticeably thinner for the LEWICE ice shape compared to the other two ice shapes, showing less momentum transport due to turbulence from the flow exposed surface of the wing. It is interesting to note the higher magnitude Reynolds stress closer to the lower surface leading edge for both the LEWICE and full-scale compared to the scaled ice shape at  $-12^\circ$  angle of attack. The stronger Reynolds stress could indicate the presence of smaller coherent length scales for the full-scale and LEWICE ice shapes compared to the scaled ice shape. This will be confirmed by the POD modal analysis results.

#### D. Reynolds Stress: $n_0 = 0.5$ Ice Accretions

Figure 12 includes the Reynolds stress contours for the  $n_0 = 0.5$  ice shapes. It is important to note that the contour range differs between the on-body and free shear layer contours. The contour range for the on-body contours



is provided on the left-hand side of Figure 12 and the contour range for the free shear layer contour is provided on the right-hand side of Figure 12. As was the case for the first set of ice shapes, the Reynolds stress contours showed similar trends as found in the normalized mean velocity contours.

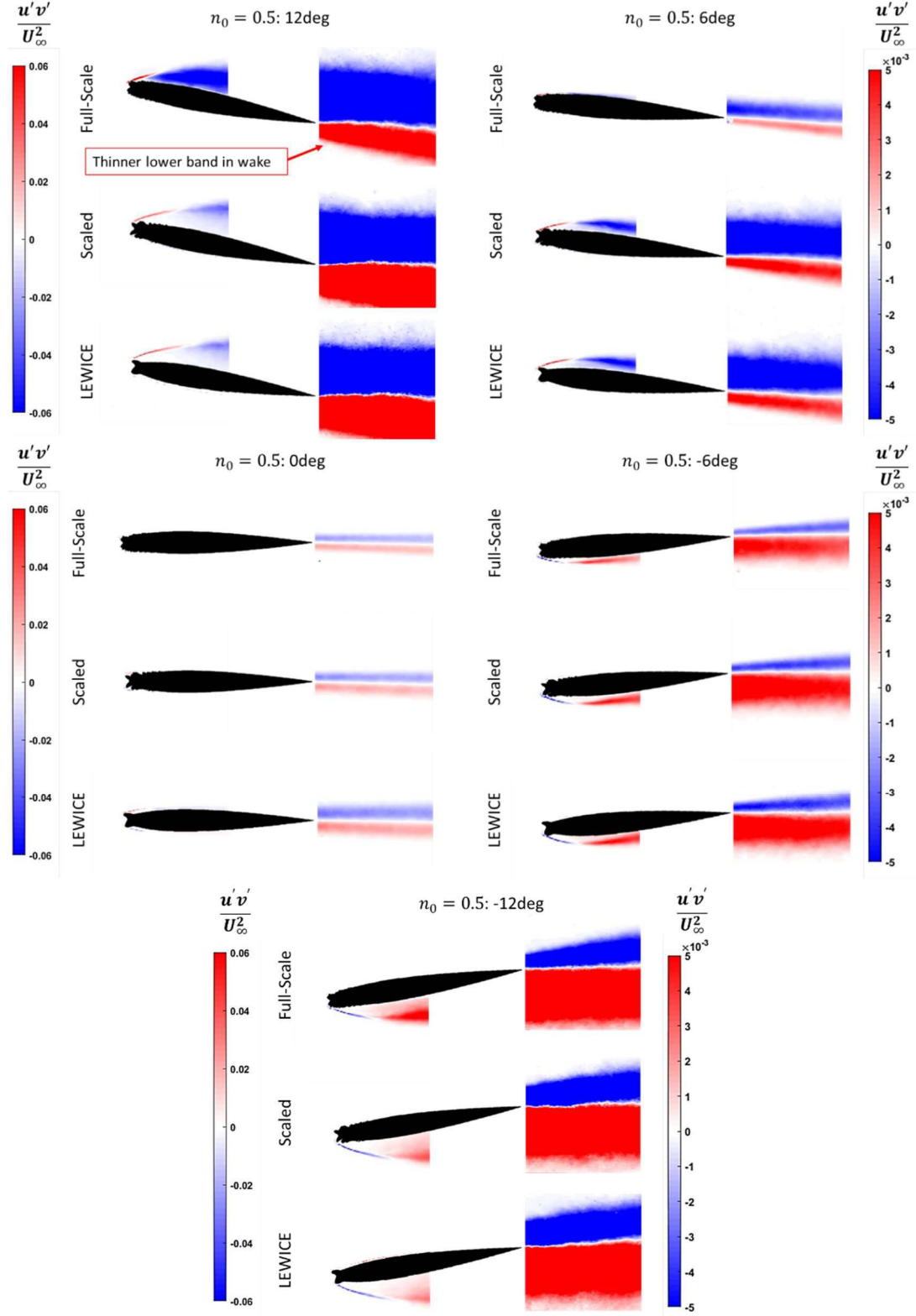


Figure 12 Reynolds stress contours for  $n_0 = 0.5$  ice shapes

A strong negative Reynolds stress is observed on the upper surface of the full-scale ice shape when compared to the other two ice shapes at  $12^\circ$ . This indicates that the flow field on the upper surface of the full-scale ice shape is more turbulent when compared to the other two cases. All three ice shapes have Reynolds stress bands corresponding to the upper surface (blue band) of similar thickness, but the Reynolds stress band corresponding to the lower surface (red band) of the full-scale is significantly thinner than the other two ice shapes as annotated in Figure 12.

At  $6^\circ$ , the scaled and LEWICE ice shapes have a strong negative Reynolds stress on the upper surface, whereas almost no Reynolds stress is present on the upper surface of the full-scale ice shape. This corresponds with the normalized streamwise velocity contours. The near wake Reynolds stress bands of the full-scale ice shape are significantly thinner than those of the other two ice shapes. The near wake Reynolds stress magnitude is also slightly lower than the scaled and LEWICE ice shapes, showing less momentum transport due to turbulence with respect to the full-scale ice shape.

This trend continues at  $0^\circ$ , where the near wake Reynolds stress bands of the full-scale are again the thinnest. The scaled is more similar to the full-scale than the LEWICE, which has significantly thicker bands in the wake. This corresponds with the thicker wake of the LEWICE ice shape shown by the normalized velocity contours. All three ice shapes have minimal Reynolds stress on either the upper or lower surface.

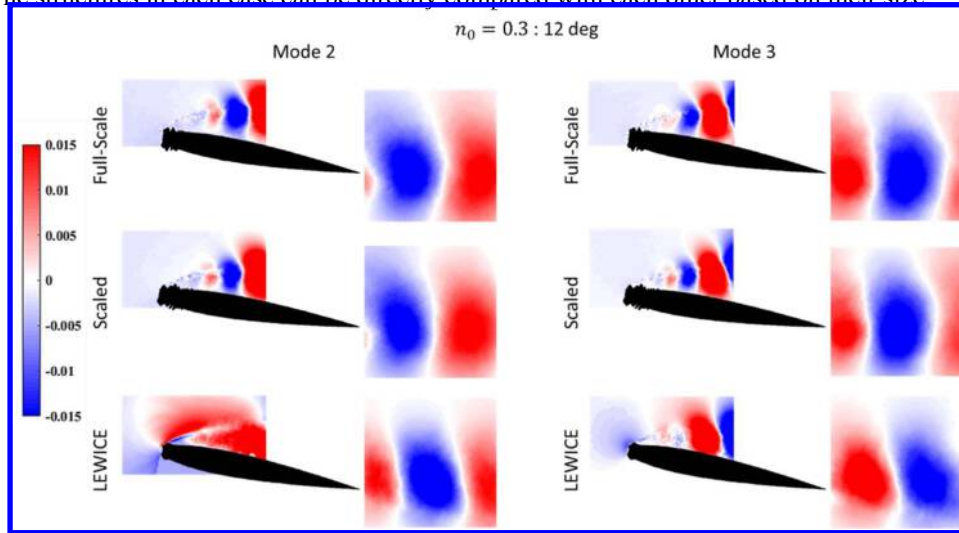
The Reynolds stress contours at  $-6^\circ$  continue to show higher Reynolds stress in the LEWICE and scaled ice shapes when compared to the full-scale ice shape. Strong positive Reynolds stress is seen on the lower surfaces and near wake of the LEWICE and scaled ice shapes. The Reynolds stress on the lower surface and in the near wake of the full-scale ice shape is lower in magnitude and thinner than those of the LEWICE and scaled ice shapes.

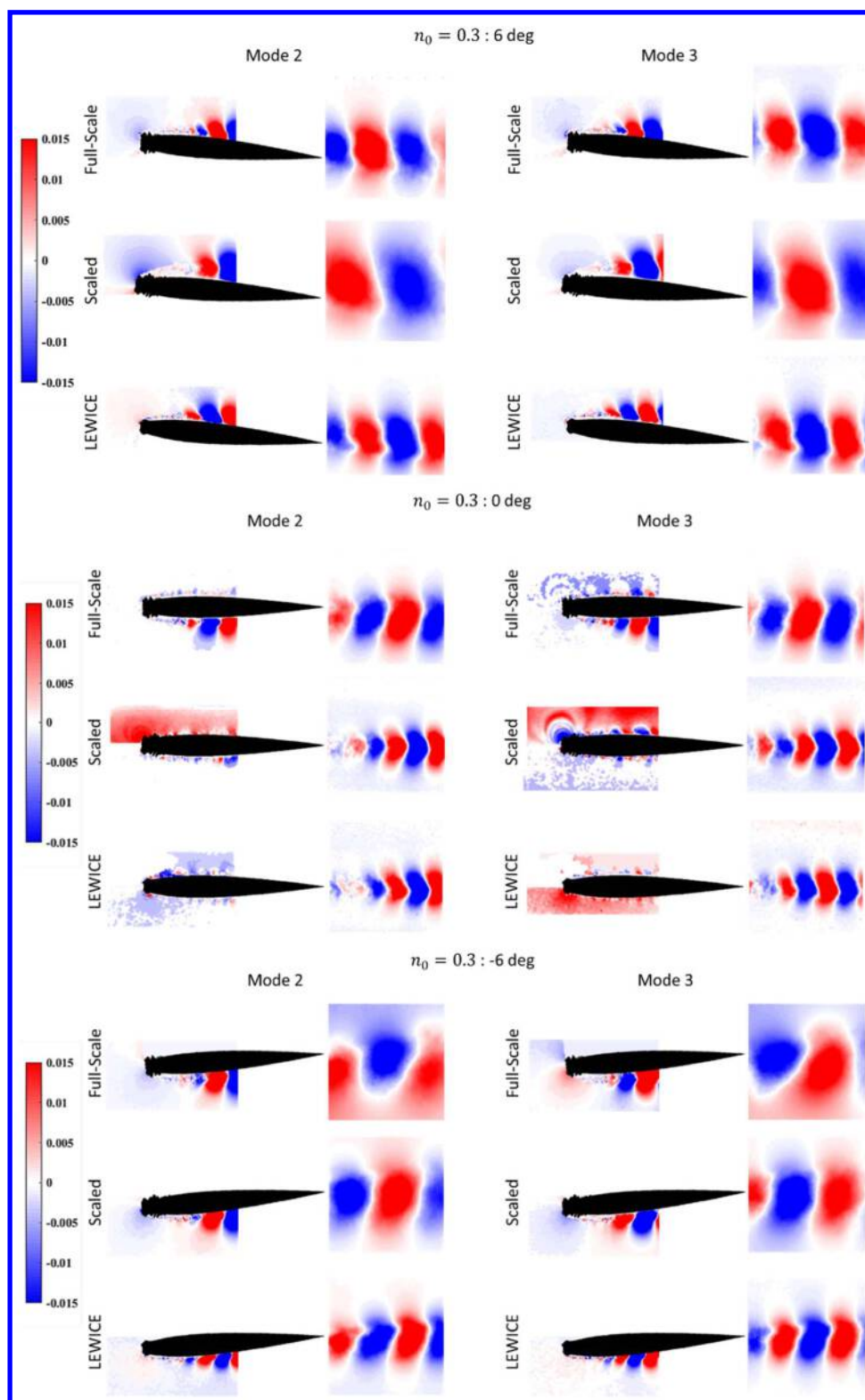
A strong Reynolds stress band corresponding to the lower surface can be seen throughout the near wake contours of all three ice shapes at  $-12^\circ$  angle of attack. The Reynolds stress corresponding to the upper surface of the full-scale ice shape is thinner than the other two cases.

## VI. POD Flow Field Analysis

### A. V-Velocity POD Modes: $n_0 = 0.3$ Ice Accretions

Figure 13 shows the second and third modes with respect to the V-velocity (wall-normal) for the  $n_0 = 0.3$  of ice shapes. As shown, shedding is present and coherent structures can be seen throughout the wake profiles for all ice shapes. The structures in each case can be directly compared with each other based on their size.





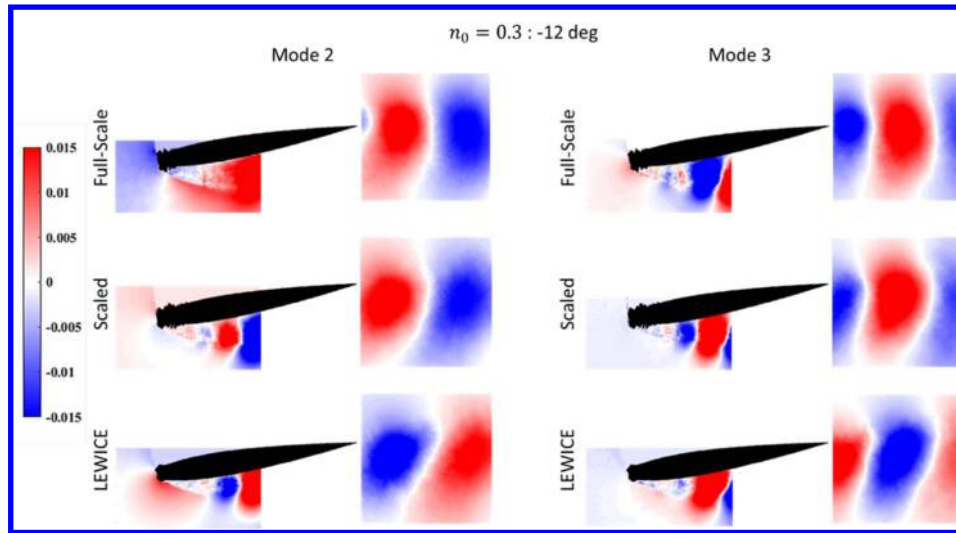


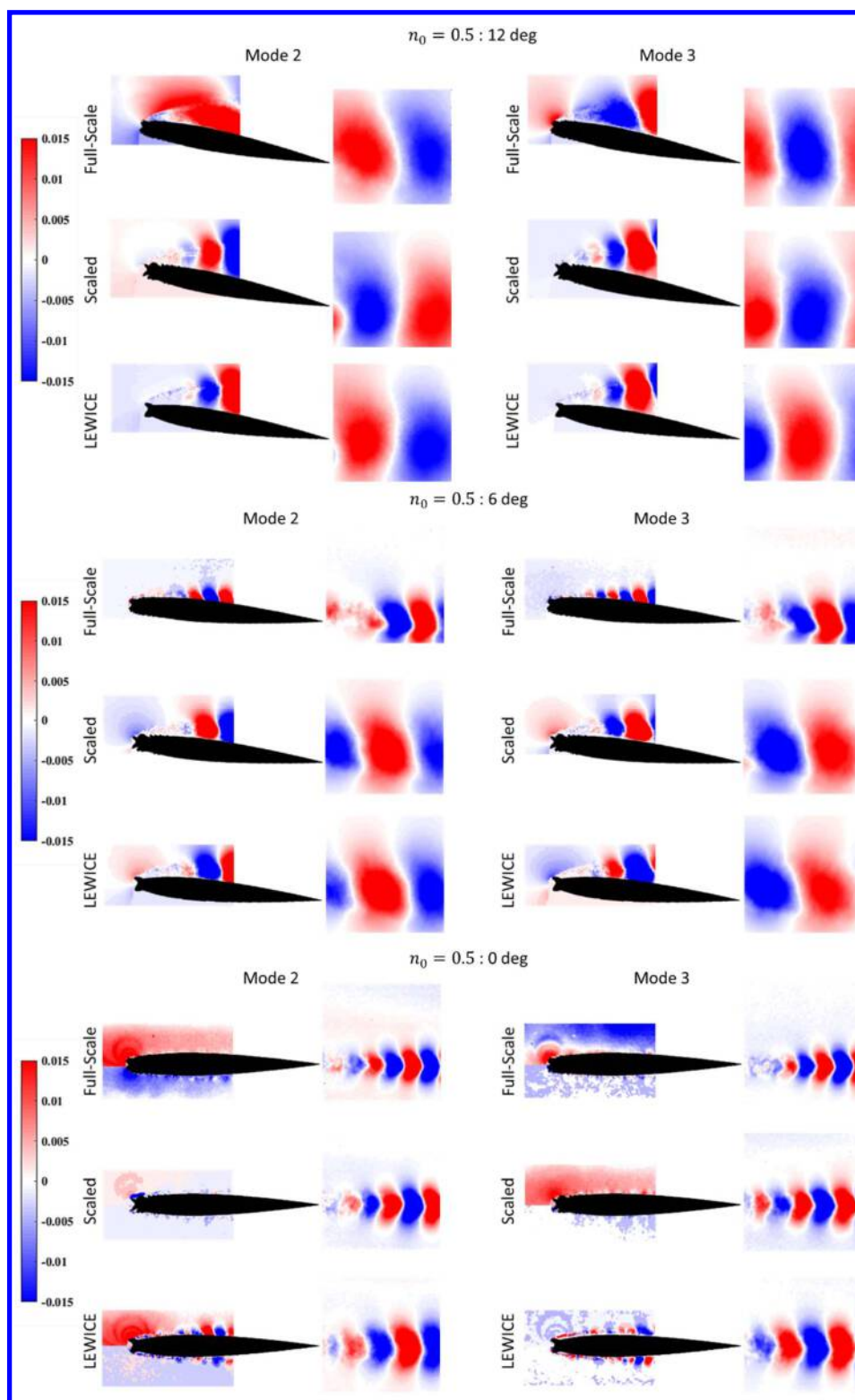
Figure 13 Second and third POD V-velocity correlation modes for  $n_0 = 0.3$  ice accretion cases

The coherent structures in the wake are almost identical between the scaled and full-scale ice shapes at positive and negative  $12^\circ$  in both the second and third modes. The on-body structures are also very similar at  $12^\circ$ , but vary in the second mode at  $-12^\circ$ . The full-scale on-body contour shows no coherent structures, whereas clearly visible structures are present on the lower surface of the full-scale in the second mode at  $-12^\circ$ . The on-body and wake V-velocity coherent structures are larger for the scaled ice shape in both modes than the full-scale at  $6^\circ$ . This is due to the comparatively large further aft positioned feather region ice accretion generating greater separation than the full-scale upper surface ice accretion. The effect of the taller lower horn structure of the full-scale can be seen in both modes in the on-body and wake coherent structures at  $0^\circ$  and  $-6^\circ$  angle of attack. At these angles of attack, the coherent structures of the full-scale ice shape, in both modes, are larger than those produced by the scaled ice shape.

When comparing the LEWICE and full-scale ice shapes, the on-body structures vary in the second mode at  $12^\circ$  angle of attack, where the LEWICE shows no on-body structures. Though the wake structures are similar in size to the full-scale wake structures at  $12^\circ$  angle of attack, the LEWICE flow field is dominated by smaller structures, as indicated by the increase in the length scale of the coherent structures between the second and third modes. The full-scale flow field, on the other hand, is dominated by larger structures at  $12^\circ$  angle of attack. The LEWICE compares well to the full-scale at  $6^\circ$ , though having slightly smaller structures in the second mode than the full-scale. At  $0^\circ$  and  $-6^\circ$  angles of attack, the LEWICE produces significantly smaller on-body and wake structures than the full-scale ice shape. The V-velocity structures in the wake are very similar between the LEWICE and full-scale at  $-12^\circ$ , but vary with respect to the on-body in the second mode. The LEWICE ice shape shows on-body structures in the second mode at  $-12^\circ$  AOA, whereas the full-scale does not.

### B. V-Velocity Correlations: $n_0 = 0.5$ Ice Accretions

Figure 14 shows the second and third modes with respect to the V-velocity (wall-normal) of ice shapes collected at  $n_0 = 0.5$ . As shown, shedding is present and coherent structures can be seen throughout the wake profiles for all ice shapes. The structures are easily compared based on size. The V-velocity modes show strong similarity between the LEWICE and scaled ice accretions, as seen in the normalized streamwise velocity contours and the Reynolds stress contours.





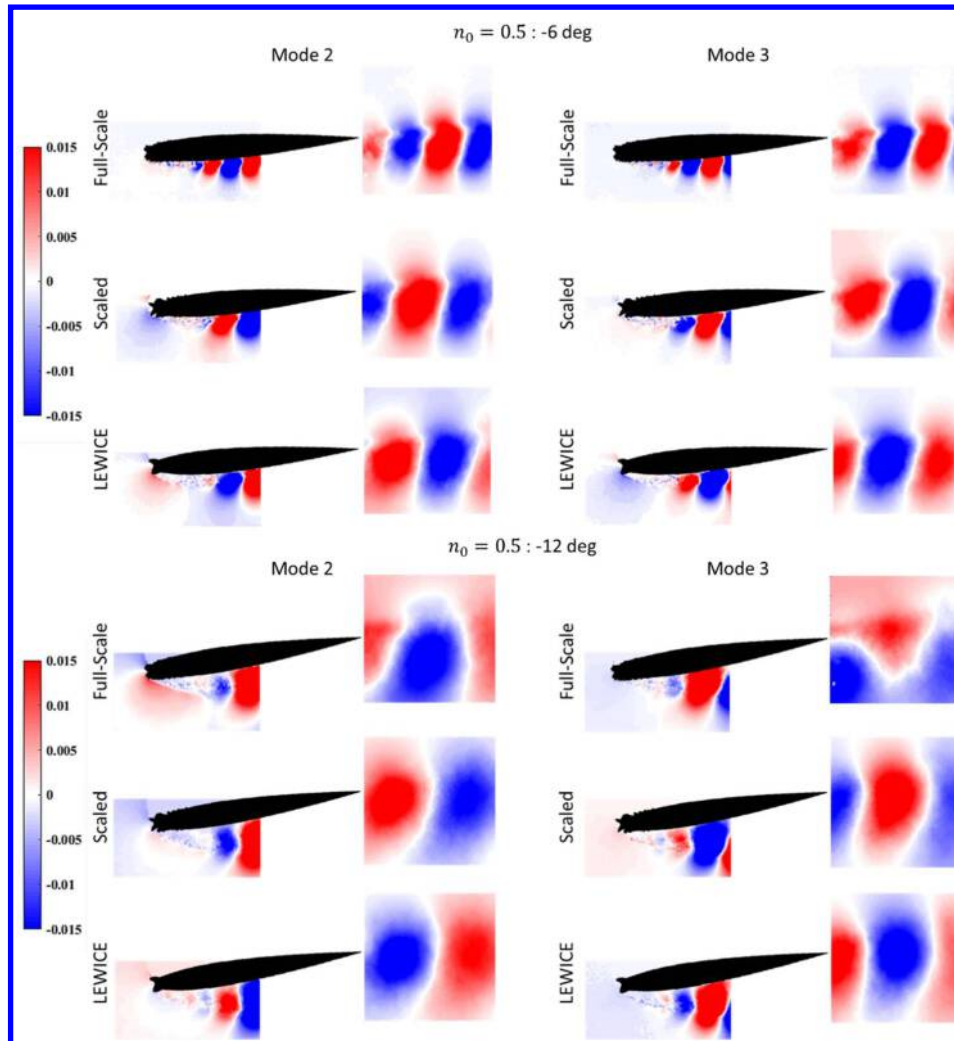


Figure 37: Second and third POD V-velocity correlation modes for  $n_0 = 0.5$  ice accretion cases

The structures in the near wake at  $12^\circ$  angle of attack are similar for both modes for all three ice shapes. However, the significant differences exist in the boundary layer. The coherent structures shown in the second and third modes have larger length scales than the other two cases. The LEWICE and scaled ice shapes both generate structures which are similar in size. This trend continues at  $6^\circ$ , where the full-scale ice shape produces smaller coherent structures at the same energy level both on the surface and in the wake when compared to the other two cases.

As expected, based on the greater Reynolds stress magnitude, the LEWICE ice shape has the largest coherent structures of the three ice shapes at  $0^\circ$  and  $-6^\circ$  angles of attack. For a given energy level, the full-scale ice shape produces coherent structures with shorter length scales than the LEWICE and scaled ice shapes at these angles of attack. At  $-12^\circ$  the size of the on-body structures for all three ice shapes are comparable for both the second and third modes.

## VII. Conclusions

### A. Ice Shape Scaling Method Aerodynamic Verification Conclusion

The results showed inconsistency in the ability of the Anderson and Tsao [3] recommended ice shape scaling method of producing an ice accretion which maintains the aerodynamics and the associated flow physics with that of a full-scale ice shape. For the ice accretions collected with a stagnation freezing fraction of  $n_0 = 0.3$ , the scaled ice shape showed very similar behavior as the full-scale ice shape, except for small deviations at the midrange positive angles of attack. It was clear that the large feather region on the upper surface of the scaled ice shape was the main cause of the deviation from the full-scale at the midrange positive angles of attack. The mean streamwise velocity and Reynolds stress contours around the leading-edge showed a clear separation point at the tip of the upper surface feather

region ice accretion causing the deviation from the full-scale ice shape. Overall the scaled ice shape collected with a  $n_0 = 0.3$  was mostly characteristic of the corresponding full-scale ice shape.

This agreement between the full-scale and the scaled ice shape was not observed in the  $n_0 = 0.5$  cases. The similar lower horn geometries of the scaled and full-scale ice shapes produced similar flow physics at negative angles of attack. However, this was not the case at positive angles of attack, where the geometric variation of the upper horns caused the aerodynamics between the scaled and full-scale ice shapes to vary. The scaled ice shape also showed earlier flow separation at both positive and negative angles of attack. Overall, the scaled ice shape collected with a  $n_0 = 0.5$  was uncharacteristic of the corresponding full-scale collected ice shape.

The difference in the evaluation of the scaled and full-scale ice shapes for the SLD conditions evaluated in this research show inconsistency in the ability of the Anderson and Tsao [3] recommended ice shape scaling method to reproduce an ice shape aerodynamically characteristic of the full-scale ice accretion in SLD icing conditions. Therefore, it can be concluded that the ice accretion generated by the scaling method does not deliver the same aerodynamic effect as the full-scale ice shape. Rather, the results suggest a possible stagnation freezing fraction constraint for the scaling method's use in SLD conditions, as this was the primary icing collection parameter difference between the ice shapes collected with a  $n_0 = 0.3$  and  $n_0 = 0.5$ . The results also expressed the necessity for better preservation of the feather region ice accretion geometry and frontal horn height, as these provided the largest deviations in the aerodynamics of the different ice shapes. This along with the inconsistency of the scaling method to produce an ice shape that preserved the associated aerodynamics with the full-scale ice accretion between the two stagnation freezing fraction cases suggest the necessity of further refinement of the recommended ice shape scaling method for SLD conditions.

### C. LEWICE Software Aerodynamic Verification Conclusion

Unlike in the case of the scaling method, the LEWICE predicted ice shapes were consistently uncharacteristic of the full-scale ice accretion. For the LEWICE ice shape predicted based on the icing parameters of the full-scale ice accretion collected with a  $n_0 = 0.3$ , the LEWICE ice shape was aerodynamically uncharacteristic of the full-scale ice shape throughout the angle of attack range, with a small exception at midrange positive angles of attack. The LEWICE ice shape consistently produced less transfer of momentum due to turbulence in the flow and produced smaller coherent structures. Overall, the LEWICE ice shape simulated with a  $n_0 = 0.3$  was uncharacteristic of the corresponding full-scale collected ice shape.

The LEWICE predicted ice shape with a  $n_0 = 0.5$  was also uncharacteristic of the corresponding full-scale ice shape. Though showing some aerodynamic similarity to the full-scale at negative angles of attack, the modal analysis deviation at this range between the two ice shapes. For the rest of the angle of attack range, the LEWICE ice shape consistently produced different flow characteristics than the full-scale ice accretion. This was expressed across all experimentation.

Overall, the LEWICE software was consistently unable to predict an ice shape aerodynamically characteristic of the full-scale ice accretion for the SLD conditions evaluated in this paper. Therefore, the LEWICE software fails to be aerodynamically verified for SLD conditions. This is expressed by the inconsistency in the under or over prediction of the full-scale icing effect on the aerodynamics. LEWICE and similar software are typically used during the aircraft design process. However, the results presented suggest that this is poor practice since the LEWICE does not consistently predict an ice shape with a similar or greater effect on the associated aerodynamics, as shown by the  $n_0 = 0.3$  case. This could result in a non-conservative approach during the aircraft design process and could lead to the necessity of redesign following experimental evaluation. The LEWICE software is still useful for studying icing with respect to the main ice formation, but experimentation is still required for studying SLD conditions.

## VIII. Future Work

The verification performed in this thesis focused on two-dimensional ice geometries collected through tracing the ice shape at the model midspan for the scale and full-scale models in the NASA IRT and predicted through the two-dimensional prediction condition of the LEWICE software. NASA Glenn IRT as the ability to record ice shapes three-dimensionally, therefore, further work is needed to assess the effect of the spanwise variation of the ice shape on the aerodynamic verification of the scaling method and LEWICE three-dimensional predictions. The spanwise variation may cause better agreement between the scaled and full-scale ice shapes, if the geometric differences are the most extreme at midspan location where the two-dimensional tracings were collected, however this is unknown. The research presented provides a template for experimentally evaluating the aerodynamics of different ice accretions, allowing for further SLD conditions and of three-dimensionally recorded and predicted ice accretions to be studied. This in turn will allow for verification of ice shape scaling methods and ice shape predictive software to be assessed

for aerodynamic verification. It is important to note that multiple spanwise positions will need to be evaluated for three-dimensional ice shapes to fully understand the effect of the spanwise variation in the ice shapes and the to fully assess the aerodynamic verification based on three-dimensional ice accretions.

## Appendix A

### LEWICE Ice Shape Input Files Based on Full-Scale Icing Parameters

#### $n_0 = 0.3$ LEWICE ice shape input file:

```
03_24_05_Run01
&LEW20
TSTOP = 1374
IBOD = 1
IFLO = 1
RHOP = 998.17
SLD = 1
&END
&DIST
FLWC = 0.083237004, 0.088925623, 0.061741051, 0.041678313, 0.03930365, 0.035597823, 0.036735779,
0.037244728, 0.036774058, 0.045125665, 0.045653839, 0.04300063, 0.035311774, 0.030850271, 0.032089732,
0.02956732, 0.090740401, 0.090182803, 0.096239536
DPD = 16.65236861, 38.84918941, 65.12431469, 82.5, 97.5, 112.5, 127.5, 142.5, 157.5, 172.5, 187.5, 202.5, 217.5,
232.5, 247.5, 262.5, 297.1906713, 388.176182, 578.0622573
&END
&ICE1
CHORD = 1.83
VINP = 51.481
LWC = 1.45
TINF = 261.15
SREF = 1
&END
&LPRNT
&END
&RDATA
&END
&BOOT
&END
```

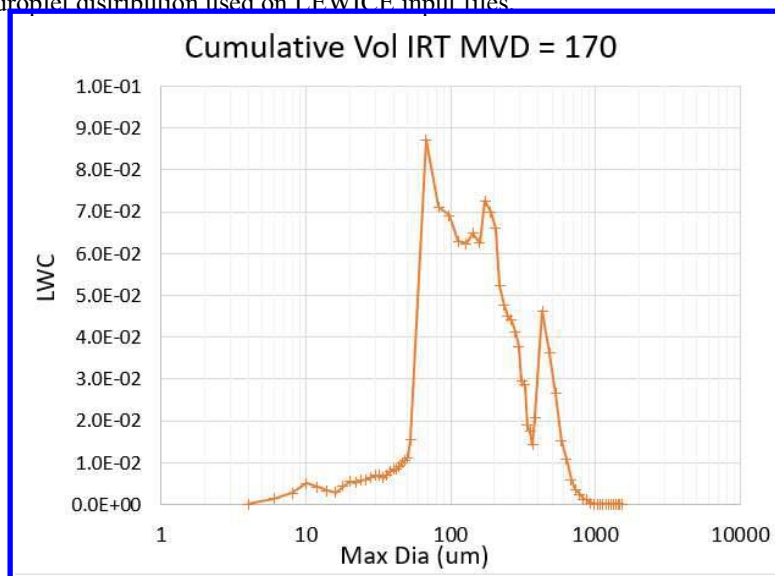
#### $n_0 = 0.5$ LEWICE ice shape input file:

```
03_24_05_Run03
&LEW20
TSTOP = 1374
IBOD = 1
IFLO = 1
RHOP = 993.55
SLD = 1
&END
&DIST
FLWC = 0.083237004, 0.088925623, 0.061741051, 0.041678313, 0.03930365, 0.035597823, 0.036735779,
0.037244728, 0.036774058, 0.045125665, 0.045653839, 0.04300063, 0.035311774, 0.030850271, 0.032089732,
0.02956732, 0.090740401, 0.090182803, 0.096239536
DPD = 16.65236861, 38.84918941, 65.12431469, 82.5, 97.5, 112.5, 127.5, 142.5, 157.5, 172.5, 187.5, 202.5, 217.5,
232.5, 247.5, 262.5, 297.1906713, 388.176182, 578.0622573
&END
&ICE1
```

CHORD = 1.83  
 VINFL = 51.481  
 LWC = 1.45  
 TINFL = 252.15  
 SREF = 1  
 &END  
 &LPRNT  
 &END  
 &RDATA  
 &END  
 &BOOT  
 &END

## Appendix B

MVD 170µm NASA Glenn Icing Research Tunnel Droplet Distribution Data  
 Cumulative volume droplet distribution used on LEWICE input files



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